BANKING AND FINANCIAL TRANSACTION DATA ANALYSIS

PROJECT OVERVIEW:

This project aims to analyze banking transaction data to uncover insights on spending , fraud detection, and credit risk. Power BI, We clean, transform, and visualize the data to support informed financial decision-making.

OBJECTIVE:

- Analyze financial transaction data to uncover patterns in customer behavior and bank operations.
- Identify anomalies or suspicious transactions for fraud detection.
- Evaluate credit risk using loan and income data.
- Visualize key metrics like monthly spending trends, transaction types, account balances, and loan approvals.
- Empower financial decision-making through an interactive, insight-driven power bi dashboard

DATA SOURCE & DESCRIPTION

SOURCE:

The dataset is taken from **kaggle** and simulates real-world data from various banking transactions data in india.

DESCRIPTION:

Customer information – ID, age, gender, income, region

- **Transaction records** – date, amount, type (debit, credit), merchant category
- **Account details** – account type, balance history
- **Loan data** – loan amount, approval status, repayment duration
- **Fraud indicators** – flags for suspicious or fraudulent transactions
- This data provides a comprehensive view of customer activities and financial operations, enabling meaningful analysis and insights.

DATA PREPROCESSING (EXCEL)

Data cleaning was performed in **excel** to prepare the dataset for analysis and visualization.

The following steps were taken:

✅ **Removed duplicates** to ensure data consistency

✅ **Handled missing values** by filling or removing null entries

✅ **Formatted dates and currency fields** for consistency and readability

✅ **Created derived columns** to enhance analysis, such as:

- *Monthly spending*
- *Loan-to-income ratio*

SQL QUERIES (MYSQL)

To extract and analyze key insights, the following SQL operations were performed:

- **Data import:** loaded the cleaned excel data into mysql tables.

  Filtering – Filtered data by transaction type, amount, region, and fraud flags
  Aggregation – Calculated totals, averages, and trends
  Joins – Linked customer, transaction, loan, and account tables
  Grouping – Grouped data by region, account type, and customer
  Ordering – Sorted customers by spending, loan approvals, or fraud likelihood.
- Grouped customers by region to identify distribution and *sorted customers by highest total spending.*

POWER BI DASHBOARD:

An interactive power BI dashboard was created to visualize institutional performance metrics clearly and effectively.

Key features:

- 📊 **Spending insights**: track total and average spending per customer across months and transaction types
  💳 **transaction analysis**: visualize credit vs. Debit transactions and their volume over time
  🏛️ **account overview**: monitor balances by account type and customer segment
  💰 **loan insights**: view loan approval status, amounts, and loan-to-income ratios
  🚨 **fraud detection**: highlight suspicious transactions and patterns for further review
  📈 **trends & filters**: use slicers to filter data by customer, region, transaction type, and time range
  🧠 **decision-making support**: the dashboard empowers banks to enhance services, reduce fraud, and improve credit risk assessment

CONCLUSION:

This project demonstrated the power of data analytics in transforming raw banking data into actionable insights. By leveraging tools like mysql and power BI, we successfully:

- Identified patterns in customer spending and account behavior
- Detected anomalies and potential fraud
- Evaluated loan risk based on income and approval data
- Enabled clear, data-driven decision-making for banking operations
- Our interactive dashboard equips stakeholders with the insights needed to enhance service delivery, manage risk, and improve customer satisfaction in a dynamic financial environment.

THANK YOU